

Transfer Learning Through Conditional Quantile Matching

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Motivation

Problem: Target-domain labeled data are **scarce**; source-domain data are abundant but **shifted**.

We observe the following two categories of data:

- ▶ Target-domain data $\mathcal{D}_T = \{(X_i^{(0)}, Y_i^{(0)})\}_{i=1}^{n_0} \sim P^{(0)}$.
- ▶ K source-domain data $\mathcal{D}_S^{(k)} = \{(X_i^{(k)}, Y_i^{(k)})\}_{i=1}^{n_k} \sim P^{(k)}$ for $k = 1, \dots, K$.

Two standard assumptions in the transfer learning literature:

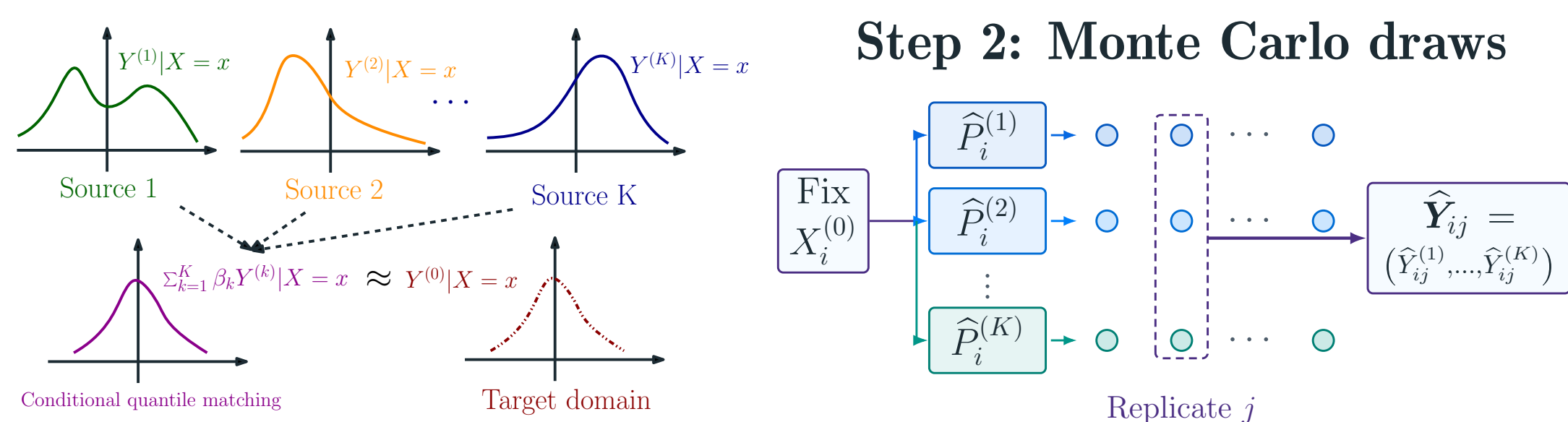
- ▶ *Covariate Shift:* $P^{(k)}(y|x) = P^{(0)}(y|x)$ but $P^{(k)}(x) \neq P^{(0)}(x)$;
- ▶ *Label Shift:* $P^{(k)}(x|y) = P^{(0)}(x|y)$ but $P^{(k)}(y) \neq P^{(0)}(y)$.

Synthetic Data Augmentation: Generate high-quality labeled data for the target domain when, for all $k = 1, \dots, K$,

$$P^{(0)}(x) \neq P^{(k)}(x) \quad \text{and} \quad P^{(0)}(y|x) \neq P^{(k)}(y|x).$$

TLCQM Pipeline

Main Idea: Learn source conditional generators, generate synthetic responses at target covariates, then calibrate them by conditional quantile matching.



Step 1: Learn a generator $\hat{P}^{(k)}(y|x)$ for each source domain $\mathcal{D}_S^{(k)}$.

Step 2: For target covariates $X_i^{(0)}$, draw synthetic response vectors $\hat{\mathbf{Y}}_{ij} = (\hat{Y}_{ij}^{(1)}, \dots, \hat{Y}_{ij}^{(K)})$, $j = 1, \dots, M$ with $\hat{Y}_{ij}^{(k)} \sim \hat{P}^{(k)}(y|X_i^{(0)})$.

Step 3: Compute the conditional quantile calibration estimator

$$\hat{\beta} \in \arg \min_{\beta \in \Theta} \frac{1}{n_0} \sum_{i=1}^{n_0} \left\{ \frac{1}{M} \sum_{j=1}^M \left| Y_i^{(0)} - \beta^T \hat{\mathbf{V}}_{ij} \right| - \frac{1}{2M(M-1)} \sum_{j,j'=1}^M \left| \beta^T \hat{\mathbf{V}}_{ij} - \beta^T \hat{\mathbf{V}}_{ij'} \right| \right\},$$

where $\hat{\mathbf{V}}_{ij} = (1, \hat{\mathbf{Y}}_{ij})^T \in \mathbb{R}^{K+1}$.

Step 4 (optional): Estimate $w_k(x) = \frac{dP^{(0)}(x)}{dP^{(k)}(x)}$ for covariate shift.

Step 5: Obtain the augmented target data $\mathcal{D}_T \cup \mathcal{D}_A$, where $\mathcal{D}_A = \bigcup_{k=1}^K \left\{ (X_i^{(k)}, \hat{\beta}^T \hat{\mathbf{V}}_i^{(k)}) \right\}_{i,k}$ with $\hat{\mathbf{V}}_i^{(k)} = (1, \hat{Y}_i^{(1,k)}, \dots, \hat{Y}_i^{(K,k)})^T$ and $\hat{Y}_i^{(j,k)}$ as any predicted value for $Y_i^{(j)}$ from $\hat{P}^{(j)}(y|X_i^{(k)})$.

Theory: Excess Risk Decomposition

With rich source data, TLCQM moves the leading ERM complexity term from n_0 to $N = \sum_{k=0}^K n_k$, while exposing the cost of imperfect transfer.

- ▶ Given a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$, the population risk of a prediction function $f \in \mathcal{F}$ is $R(f) := \mathbb{E}_{P^{(0)}} [\ell(Y^{(0)}, f(X^{(0)}))]$.
- ▶ $\text{Rad}_n(\mathcal{F})$ denotes the Rademacher complexity of $\mathcal{F}$.
- ▶ Define the conditional continuous ranked probability score (CRPS) as

$$\hat{\Delta}(\hat{\beta}) := \mathbb{E}_{X^{(0)}} \left\{ \int_{\mathbb{R}} \left[F^{(0)}(t|X^{(0)}) - \hat{G}_{\hat{\beta}}(t|X^{(0)}) \right]^2 dt \right\}.$$

Target-only Excess Risk: Let $\hat{f}^{(0)}$ be the empirical risk minimizer (ERM) under target-only data:

$$R(\hat{f}^{(0)}) - R(f^{(0)}) \lesssim \text{Rad}_{n_0}(\mathcal{F}) + \sqrt{\frac{\log(2/\delta)}{n_0}} \quad \text{w.p.} \geq 1 - \delta.$$

TLCQM Excess Risk: Define the TLCQM ERM as $\hat{f}^{(0,tl)} = \arg \min_{f \in \mathcal{F}} \frac{1}{N} \left\{ \sum_{i=1}^{n_0} \ell(Y_i^{(0)}, f(X_i^{(0)})) + \sum_{k=1}^K \sum_{i=1}^{n_k} \hat{w}_k(X_i^{(k)}) \cdot \ell(\hat{\beta}^T \hat{\mathbf{V}}_i^{(k)}, f(X_i^{(k)})) \right\}$.

For squared loss, a closed convex $\mathcal{F} \subset L_2(P_X^{(0)})$, and the conditional mean pseudo-labels from Step 4,

$$R(\hat{f}^{(0,tl)}) - R(f^{(0)}) \lesssim \underbrace{\text{Rad}_N(\mathcal{F}) + \sqrt{\frac{\log(1/\delta)}{N}}}_{\text{Standard generalization error}} + \underbrace{\frac{1}{N} \sum_{k=1}^K \|\hat{w}_k - w_k\|_1}_{\text{Importance weight error}} + \underbrace{\frac{(N - n_0)^2 \cdot \hat{\Delta}(\hat{\beta})}{N^2}}_{\text{Conditional transfer error}} \quad \text{w.p.} \geq 1 - \delta.$$

- ▶ The decomposition is **learner-agnostic**: transfer quality is separated from the downstream model complexity.

Conditional Quantile Calibration Error

Let $\beta_* \in \arg \min_{\beta \in \Theta} \Delta(\beta)$, where $\Delta(\beta)$ is the population conditional CRPS discrepancy.

$$\hat{\Delta}(\hat{\beta}) \lesssim \underbrace{\Delta(\beta_*)}_{\text{Transfer bias}} + \underbrace{\sum_{k=1}^K \|\hat{g}^{(k)} - g^{(k)}\|_{\infty}}_{\text{Source generator error}} + \underbrace{\sqrt{\frac{[\log n_0 + \log(1/\delta)] [\Delta(\beta_*) + \sum_{k=1}^K \|\hat{g}^{(k)} - g^{(k)}\|_{\infty}]}{n_0}}}_{\text{Localized target-sample error}} + \underbrace{\frac{\log n_0 + \log(1/\delta)}{n_0} + \epsilon_{\text{opt}} + O\left(\frac{1}{\sqrt{M}}\right)}_{\text{Optimization and Monte Carlo approximation errors}}.$$

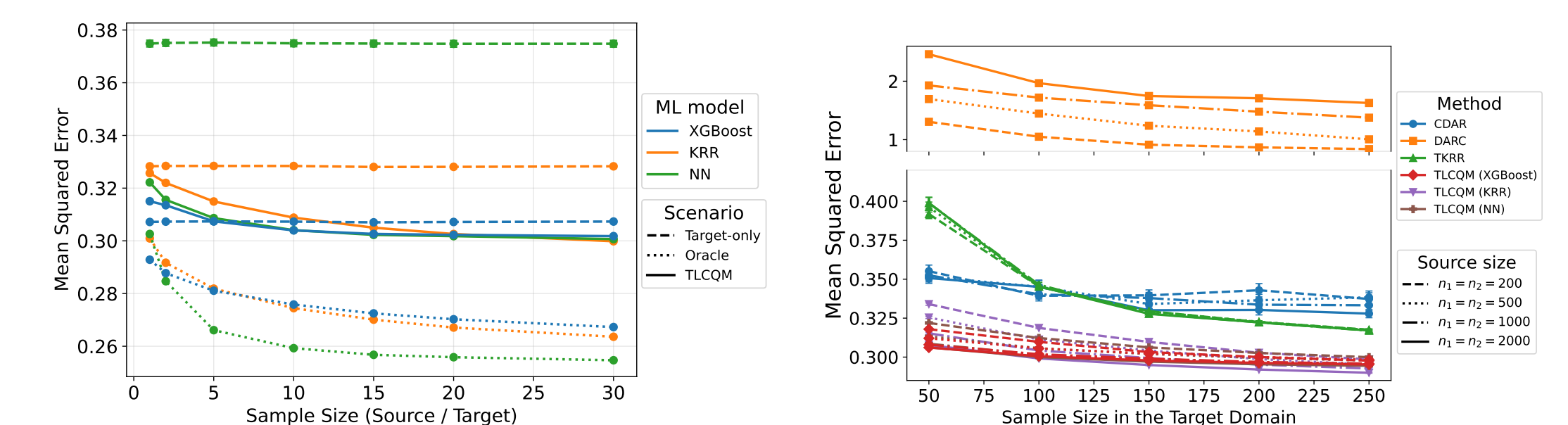
- ▶ $\Delta(\beta_*)$ is the **irreducible transfer bias**, which vanishes when the target conditional law lies in the affine span of the source conditional laws.

- ▶ Large $\Delta(\beta_*)$ can offset variance reduction, causing **negative transfer**.

Simulations: General Target Shift

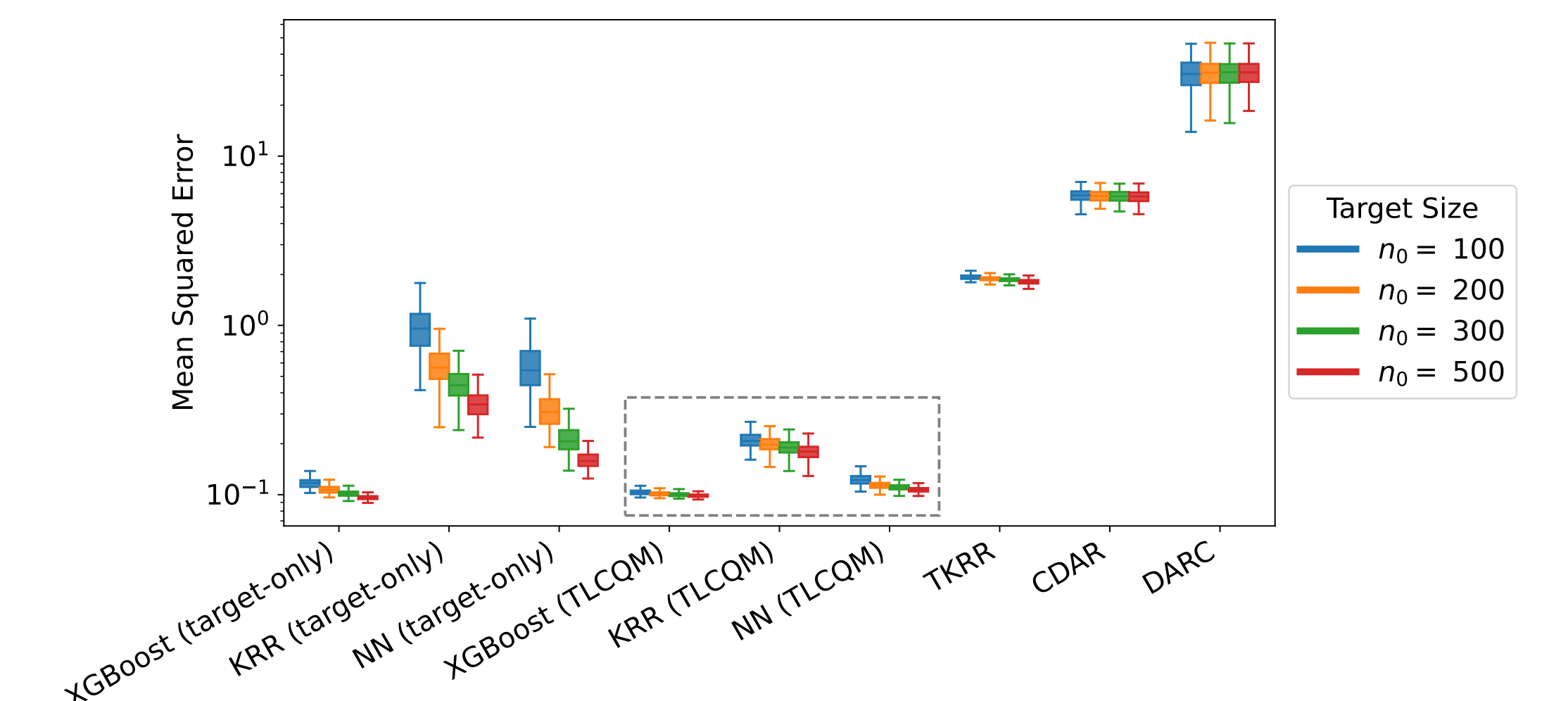
We model each source-domain conditional distribution as $g^{(k)}(x, \cdot)_{\#} P_{\eta} = P^{(k)}(\cdot|x)$, and estimate $g^{(k)}$ via engression: $\hat{g}^{(k)} \in \arg \min_{g \in \mathcal{F}} \frac{1}{n_k} \sum_{i=1}^{n_k} \left[\frac{1}{m} \sum_{j=1}^m |Y_i^{(k)} - g(X_i^{(k)}, \eta_{ij})| - \frac{1}{2m(m-1)} \sum_{j=1}^m \sum_{j'=1}^m |g(X_i^{(k)}, \eta_{ij}) - g(X_i^{(k)}, \eta_{ij'})| \right]$, where P_{η} is a pre-specified noise distribution.

- ▶ Source domains: $Y^{(1)} = \sin(3\theta^T X^{(1)}) + 1 + \epsilon^{(1)}$ and $Y^{(2)} = 2 \cos(3\theta^T X^{(2)}) + 1 + \epsilon^{(2)}$, where $\theta = (1, \frac{1}{2}, \dots, \frac{1}{5})^T \in \mathbb{R}^5$, $X^{(1)}, X^{(2)} \sim \mathcal{N}(\mathbf{1}_5, \mathbf{I}_5)$, and $\epsilon^{(1)}, \epsilon^{(2)} \sim \mathcal{N}(0, 0.25)$.
- ▶ Target domain: $Y^{(0)} = \frac{1}{3} \sin(3\theta^T X^{(0)}) - 3 + \epsilon^{(0)}$ with $X^{(0)} \sim \mathcal{N}(\mathbf{0}_5, 0.25 \cdot \mathbf{I}_5)$ and $\epsilon^{(0)} \sim \mathcal{N}(0, 0.25)$.



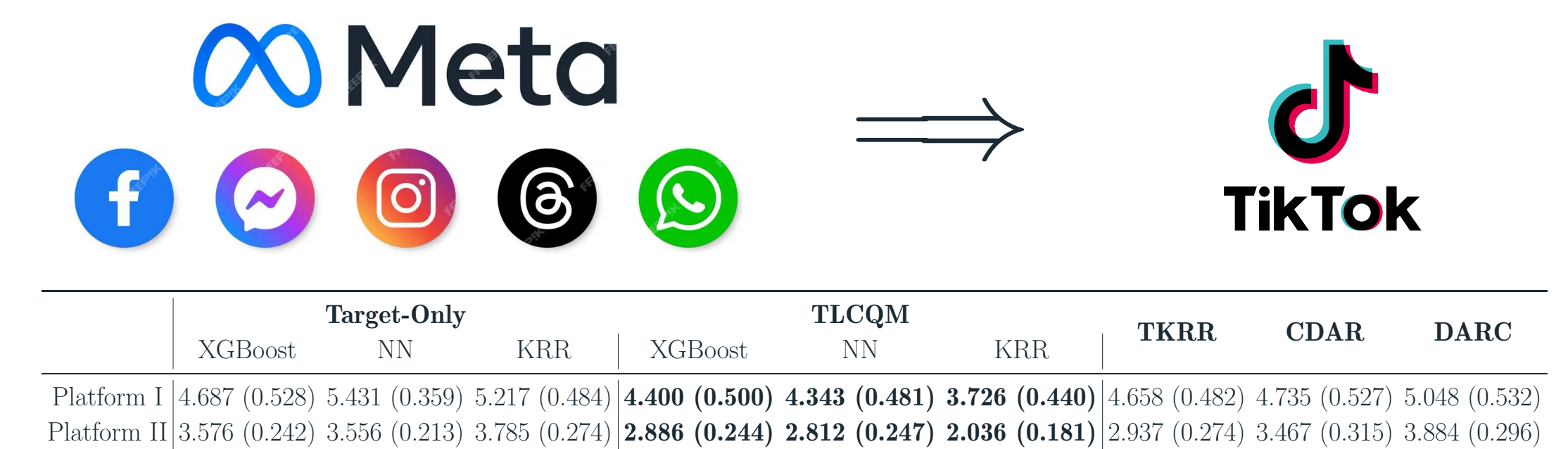
Real Data Applications

Public Apartment Data: Predict log (apartment rental price) for the target state FL using data from source states IL, OH, and WA.



- ▶ Flexible learners, especially KRR and NN, benefit most from our TLCQM framework when the target size n_0 is small.

Monthly Active User Prediction at Meta: Predict the country-level monthly active user statistics for a target app using data as from Meta's family of apps.



- ▶ TLCQM improves all three learners on both device platforms.